

# Subsystem 1 Development Guideline

## Catching Mini Games

### Subsystem 1 Description

Subsystem 1 is responsible for catching Pokémon through mini games.

It includes:

1. **PokeReflex**
2. **PokeGuess**
3. **Wild Pokémon generation**
4. **Sending catch result to Main System**

### Subsystem 1 Assignment

| Developer   | Assigned Features                               |
|-------------|-------------------------------------------------|
| Developer 1 | Player name entry and wild Pokémon generation   |
| Developer 2 | PokeReflex mini game                            |
| Developer 3 | PokeGuess mini game and Main System integration |

## **Subsystem 1 Step-by-Step Development**

### **Step 1: Create Player Name Entry**

Subsystem 1 should also ask for the player name.

**Flow:**

**Open Subsystem 1**

↓

**Enter Player Name**

↓

**Confirm Player Name**

↓

**Select Catching Mini Game**

**Example:**

**Enter Player Name:**

**Jods**

**You are playing as Jods.**

**[Continue]**

**Important:**

**The player name must match the player name used in the Main System.**

## Step 2: Create Mini Game Selection Page

The player should choose which catching mini game to play.

Options:

☐ PokeReflex

☐ PokeGuess

Flow:

Enter Player Name



Select Mini Game



Generate Wild Pokémon



Start Game

## Step 3: Generate Wild Pokémon

Before the game starts, the system should generate a wild Pokémon.

Example:

A wild Squirtle appeared!

Wild Pokémon data should include:

pokemon\_name

type

region

image

difficulty

Suggested difficulty levels:

| Difficulty | Description               |
|------------|---------------------------|
| Easy       | Common or starter Pokémon |
| Medium     | Less common Pokémon       |
| Hard       | Rare Pokémon              |

## **Step 4: Build PokeReflex**

**PokeReflex is a reaction-based catching game.**

**Game concept:**

**A target word appears on the screen.**

**The player must click the matching symbol.**

**Example (but not limited to):**

**Target: Fire Type**

**Choices:**

**[Water Symbol]**

**[Fire Symbol]**

**[Leaf Stone]**

**[Poké Ball]**

**Correct answer:**

**Fire Symbol**

**PokeReflex flow:**

**Wild Pokémon appears**

↓

**Target word appears**

↓

**Choices appear**

↓

**Player clicks answer**

↓

**System checks if answer is correct**

↓

**If correct, Pokémon is caught**

↓

**If wrong, Pokémon fled**

↓

**Result is sent to Main System**

**PokeReflex requirements:**

**[ ] Show wild Pokémon.**

**[ ] Show target word.**

**[ ] Show answer choices.**

**[ ] Check selected answer.**

**[ ] Display result.**

**[ ] Send result to Main System.**

**PokeReflex reward rules:**

**Result**

**Reward**

**Correct**

**Pokémon caught +  
coins**

**Wrong**

**Pokémon fled + no  
coins**

**Recommended coins:**

| <b>Result</b> | <b>Coins</b> |
|---------------|--------------|
|---------------|--------------|

|                |                  |
|----------------|------------------|
| <b>Correct</b> | <b>+20 coins</b> |
|----------------|------------------|

|              |                 |
|--------------|-----------------|
| <b>Wrong</b> | <b>+0 coins</b> |
|--------------|-----------------|

## Step 5: Build PokeGuess

PokeGuess is a guessing-based catching game.

Game concept:

The system gives a category or clue.

The player guesses the correct Pokémon.

Example:

Category: Fire-type starter

Clue: I am from Paldea and I look like a crocodile.

Choices:

[Charmander]

[Tepig]

[Fuecoco]

[Squirtle]

Correct answer:

Fuecoco

PokeGuess flow:

Wild Pokémon appears

↓

System shows category or clue

↓

Player chooses an answer



↓

**System checks answer**

↓

**If correct, Pokémon is caught**

↓

**If wrong, Pokémon fled**

↓

**Result is sent to Main System**

**PokeGuess requirements:**

**[ ] Generate category or clue.**

**[ ] Show answer choices.**

**[ ] Check answer.**

**[ ] Display result.**

**[ ] Send result to Main System.**

**PokeGuess reward rules:**

| <b>Result</b>  | <b>Reward</b>                      |
|----------------|------------------------------------|
| <b>Correct</b> | <b>Pokémon caught<br/>+ coins</b>  |
| <b>Wrong</b>   | <b>Pokémon fled +<br/>no coins</b> |

**Recommended coins:**

| <b>Result</b>  | <b>Coins</b>         |
|----------------|----------------------|
| <b>Correct</b> | <b>+20<br/>coins</b> |
| <b>Wrong</b>   | <b>+0<br/>coins</b>  |

## Step 6: Send Catch Result to Main System

After each catching mini game, Subsystem 1 must send the result to the Main System.

Caught result example:

```
{  
  "playerName": "Jods",  
  "pokemonName": "Squirtle",  
  "gameName": "PokeReflex",  
  "result": "caught",  
  "levelGain": 0,  
  "coinsEarned": 20,  
  "sourceSystem": "catching_subsystem"  
}
```

Fled result example:

```
{  
  "playerName": "Jods",  
  "pokemonName": "Squirtle",  
  "gameName": "PokeReflex",  
  "result": "fled",  
  "levelGain": 0,  
  "coinsEarned": 0,  
  "sourceSystem": "catching_subsystem"  
}
```

# Subsystem 1 Final Checklist

- ☐ Player name entry
- ☐ Player name confirmation
- ☐ Mini game selection
- ☐ Wild Pokémon generation
- ☐ PokeReflex mini game
- ☐ PokeGuess mini game
- ☐ Result screen
- ☐ Send caught/fled result to Main System
- ☐ Main System updates player Pokédex
- ☐ Main System updates coins
- ☐ Main System records game history

